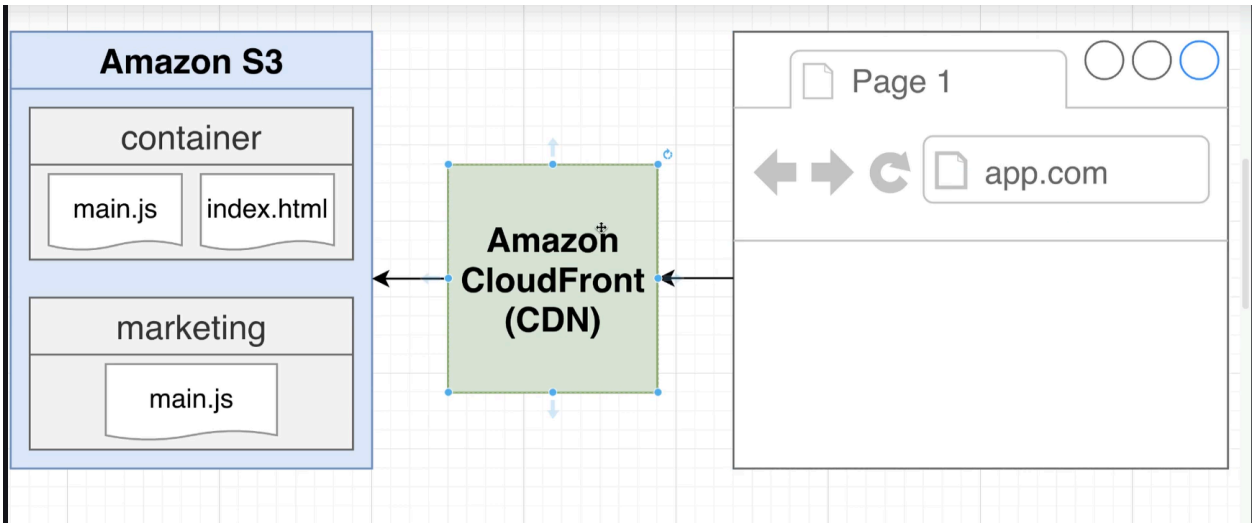
MicroFrontend Deployments

Requirements of Deployment :

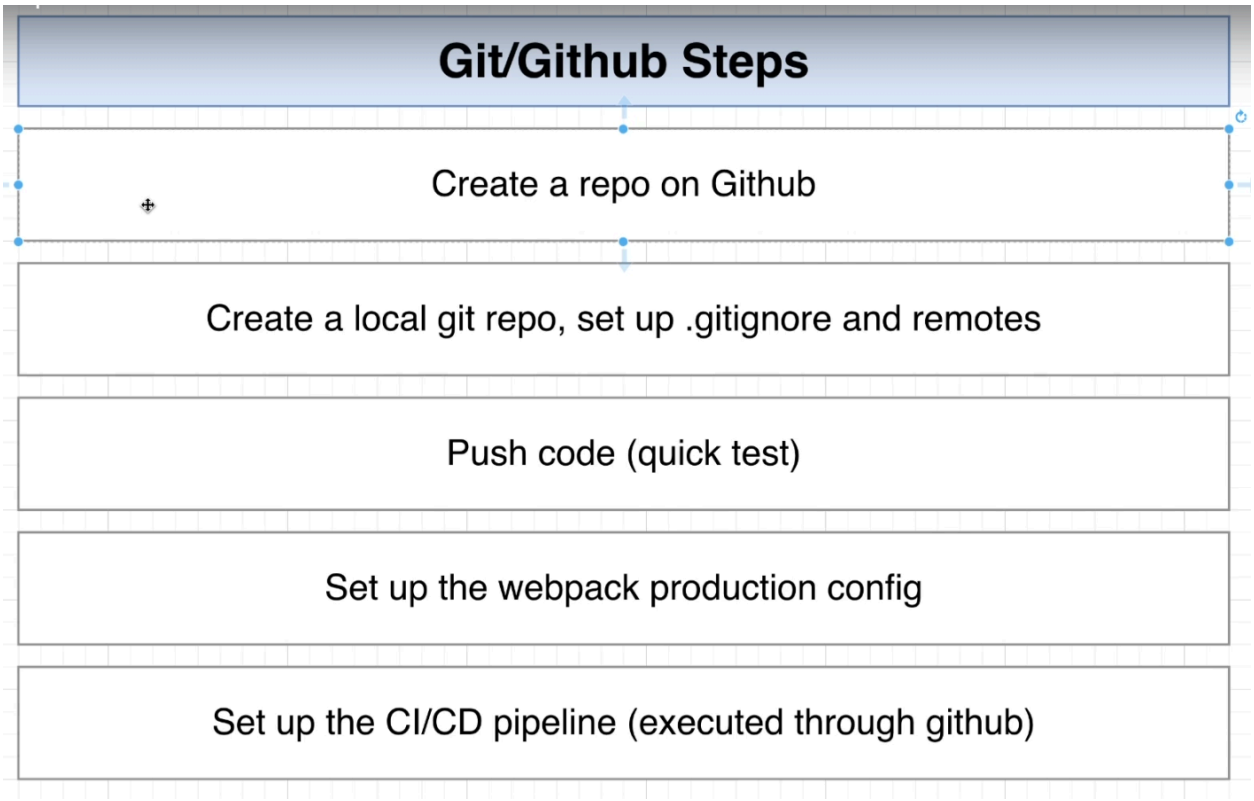
Deployment
Want to deploy each microfrontend independently (including the container)
Location of child app remoteEntry.js files must be known at *build time*!
Many front-end deployment solutions assume you're deploying a single project - we need something that can handle multiple different ones
Probably need a CI/CD pipeline of some sort
At present, the remoteEntry.js file name is fixed! Need to think about caching issues

- We will be setting up GitHub monorepo which will have multiple microfrontends. Check separately for each monorepo, whether it changed. If yes then build a production version of the microservice with webpack. Then upload the file to Amazon S3.

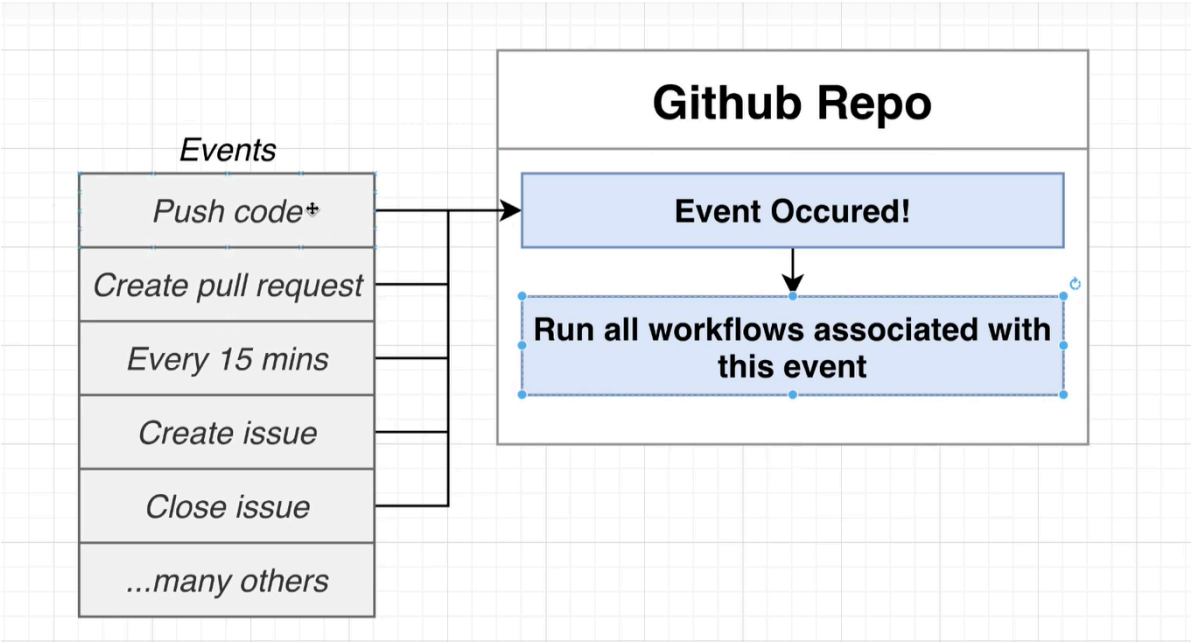


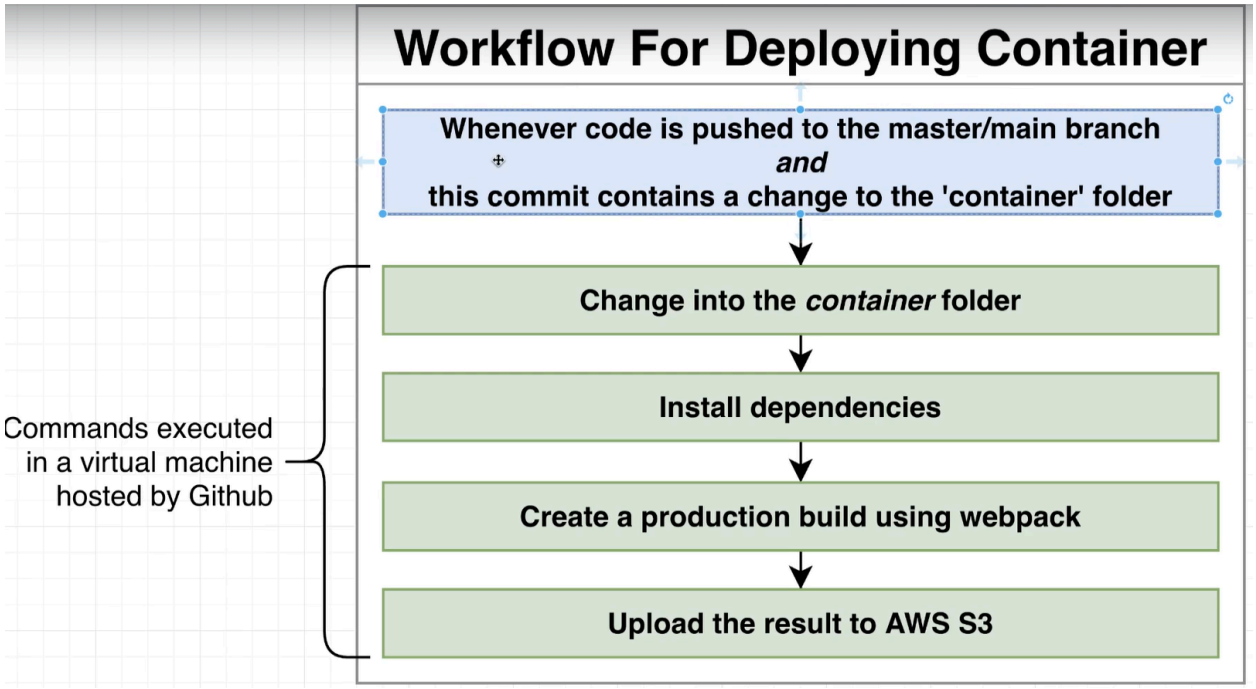
CDN stands for Content Delivery Network. A Content Delivery Network (CDN) is a geographically distributed group of servers that caches website content (images, videos, CSS, JS) at the "edge" of the internet, closer to users. It reduces latency and speeds up load times by serving data from the nearest location, rather than the origin server, while also enhancing security against DDoS attacks.

Step 1 : Git setup



- When we set mode to production in webpack file, it takes more time as it minifies the js files and make other optimizations.
- Here we make use of GitHub Actions to setup CI/CD pipeline.
    - What are GitHub Actions?





Step 2 : Create S3 bucket with unique name

Step 3 : In Properties section, enable static website hosting and add index.html.

Step 4 : In permissions section, change bucket settings to allow public access.

Step 5 : Add Bucket Policy through Permissions tab and generate it using Policy Generator

https://awspolicygen.s3.amazonaws.com/policygen.html

Add policy for GetObject and keep principle to * and add Bucket ARN of the bucket you created.

Step 6 : Create AWS CloudFront distribution. An Amazon CloudFront distribution is a configuration setting that defines how a **Content Delivery Network (CDN)** service delivers static and dynamic web content (like images, video, APIs, and HTML) to users globally.

Step 7 : Make IAM user. Follow given steps :

1. Search for "**IAM**"
2. Click "**Create Individual IAM Users**" and click "**Manage Users**"
3. Click "**Add User**"
4. Enter any name you'd like in the "**User Name**" field.
5. Click "**Next**"
6. Click "**Attach Policies Directly**"
7. Use the search bar to find and tick **AmazonS3FullAccess** and **CloudFrontFullAccess**
8. Click "**Next**"
9. Click "**Create user**"
10. Select the IAM user that was just created from the list of users
11. Click "**Security Credentials**"
12. Scroll down to find "**Access Keys**"
13. Click "**Create access key**"
14. Select "**Command Line Interface (CLI)**"
15. Scroll down and tick the "I understand..." check box and click "Next"
16. Copy and/or download the *Access Key ID* and *Secret Access Key* to use for deployment.

Step 8 : Create Invalidation in cloudfront distribution to invalidate the cache when the file is changed.

Step 9 : Add invalidation in CI/CD pipeline for index.html file.

Step 10 : Setup marketing deployment.

Step 11 : Add domain secret in github repository secrets.